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Option Hedging under Transaction Costs

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Summary

This project examines how **transaction costs** change the way options should be hedged and priced in practice. While standard financial theory assumes that options can be hedged almost perfectly through continuous trading, real markets involve execution costs, which fundamentally alter both hedging behavior and profitability. Our objective is to quantify this effect and to assess how much value is lost when simple **rule-of-thumb hedging strategies** are used instead of fully optimized ones.

We consider the case of a market maker who sells a short-dated equity call option and dynamically hedges the position using the underlying stock. Each trade incurs a small proportional cost, and the market maker is risk-averse: they care not only about average profit but also about the risk of large **hedging losses**. Using a high-frequency market simulation and an optimization algorithm, we compute the **optimal hedging strategy**. We then compare its performance to a simple **delta-hedging strategy** that rebalances at fixed time intervals, which is representative of what is often done in practice.

The results show that even small transaction costs have a meaningful impact on both **pricing** and **hedging**. For the at-the-money option studied, the fair price that compensates the market maker for risk and trading costs is noticeably higher than the standard theoretical price that ignores execution costs. In practical terms, this means that ignoring transaction costs leads to systematic **underpricing of options**.

On the hedging side, the optimal strategy does not trade continuously. Instead, it allows small mismatches between the hedge and the option exposure and only trades when these mismatches become large enough to justify the cost. This behavior leads to **state-dependent trading** rather than fixed-interval rebalancing. As a result, **hedging errors** cannot be eliminated completely: even under the optimal strategy, the final profit-and-loss distribution remains wide and slightly asymmetric, reflecting unavoidable residual risk.

When transaction costs increase, **trading frequency** declines, **hedging volatility** increases, and the option premium required by the market maker rises. When the option is far out of the money, **moneyness** plays a central role: hedging activity drops sharply and both costs and risks become much smaller.

Finally, we find that simple fixed-frequency delta hedging can perform reasonably well if the **rebalancing frequency** is carefully chosen. However, it is always dominated by the fully optimized strategy, which adapts both the size and timing of trades to market conditions. The performance gap becomes particularly pronounced when **transaction costs** are high, as rigid rebalancing either wastes money through excessive trading or exposes the trader to unnecessary risk.

Overall, the project demonstrates that transaction costs fundamentally change both how options should be hedged and how they should be priced. Optimizing hedging decisions in a **cost-aware** manner leads to materially better **risk-adjusted performance** than relying on simplified hedging rules.

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Chapter 1

Project Definition

1.1 Objective

We study the problem of hedging a short European call option in the presence of proportional transaction costs. A market maker sells the option at time 0 and dynamically trades the underlying stock over the life of the contract in order to manage the resulting exposure. At maturity, all remaining stock holdings are liquidated and the performance of the hedging strategy is evaluated.

Rather than focusing only on average profit, we measure performance through the *tracking error*, defined as the difference between the final, post-liquidation value of the hedging portfolio and the payoff of the option:

$$\mathrm{TE} = W_T - \max(S_T - K, 0), \tag{1.1}$$

where W_T denotes terminal cash wealth after liquidation and S_T is the terminal stock price.

The market maker is assumed to be risk-averse and evaluates outcomes using exponential utility:

$$\mathbb{E}^Q \left[-\exp\left(-\gamma\, \mathrm{TE}\right) \right], \tag{1.2}$$

where $\gamma > 0$ is the risk aversion parameter and expectations are taken under the risk-neutral measure Q. This formulation reflects the fact that a market maker cares not only about expected profit, but also about the risk of large hedging losses.

The primary objective of the project is to **compute and analyze the optimal dynamic hedging strategy** that maximizes this expected utility in the presence of transaction costs. A second key objective is to use the resulting value function to define a **utility-based indifference price** for selling the option, and to compare this price with the standard frictionless Black–Scholes price.

1.2 Economic Motivation

This formulation is relevant for:

- **Option market makers** who quote bid–ask spreads and must dynamically manage inventory subject to execution costs and inventory risk.
- **Buy-side investors** (e.g. funds that write covered calls or sell volatility) who warehouse option risk and rebalance discretely, incurring trading costs.

- **Utility-based pricing** frameworks where transaction costs and risk aversion generate a range of consistent option prices, rather than a single frictionless value.

The tracking error TE focuses directly on how well the hedging portfolio aligns with the option payoff net of costs, which is a natural performance measure for hedging desks.

Chapter 2

Scope Definition

Market Model

We work in the standard Black–Scholes–Merton setting under the risk-neutral measure Q:

$$\frac{dS_t}{S_t} = r\,dt + \sigma\,dZ_t^Q, \tag{2.1}$$

with constant risk-free rate r, volatility σ, and no dividends. The underlying stock S_t is the only traded risky asset. In the frictionless limit ($\lambda = 0$), a continuously rebalanced delta hedge replicates the payoffs of a European call, leading to the classical BSM price C_{BS}.

We consider a single European call with maturity T and strike K on a non-dividend paying stock, with baseline parameters:

$$T = \frac{22}{252}, \quad r = 0.05, \quad \sigma = 0.25, \quad S_0 = 100, \quad K = 100. \tag{2.2}$$

The base-case proportional transaction cost rate is

$$\lambda = 0.0005, \tag{2.3}$$

with sensitivity analysis performed later for alternative values of λ and for out-of-the-money strikes $K > 100$.

Key Tasks

Within this framework, the project addresses the following tasks:

1. Construct the CRR binomial approximation and specify portfolio dynamics with costs.
2. Solve the corresponding discrete-time dynamic programming (DP) problem and characterize the optimal hedging policy, including any no-trade region.
3. Define and compute the utility-based indifference price and its implied volatility.
4. Simulate the optimal strategy to characterize the distribution of tracking error and trading intensity.
5. Compare the optimal strategy to a heuristic Black–Scholes delta-hedging scheme with fixed rebalancing frequency.
6. Study the sensitivity of results to risk aversion γ, transaction costs λ, and strike K.

Chapter 3

Methods

3.1 Discrete-Time Approximation

We approximate the continuous-time dynamics by a Cox-Ross-Rubinstein (CRR) binomial tree with $N = 220$ time steps (10 trading opportunities per day over 22 days), so that $\Delta t = T/N$. The up and down factors are

$$u = e^{\sigma\sqrt{\Delta t}}, \qquad d = e^{-\sigma\sqrt{\Delta t}} = \frac{1}{u}, \tag{3.1}$$

and the risk-neutral probability of an up-move is

$$p = \frac{e^{r\Delta t} - d}{u - d}, \qquad 1 - p = \frac{u - e^{r\Delta t}}{u - d}. \tag{3.2}$$

At time step n, after j up-moves, the stock price is

$$S_{n,j} = S_0 u^j d^{n-j}, \qquad n = 0, \ldots, N,\ j = 0, \ldots, n. \tag{3.3}$$

3.2 Portfolio Dynamics with Transaction Costs

Let W_n denote the cash position at time t_n (in the money market account) and let θ_n denote the number of shares of the underlying stock held immediately after trading at t_n. The marked-to-market wealth before trading at time t_n is

$$\widetilde{W}_n = W_n + \theta_n S_n. \tag{3.4}$$

At time t_n, the hedger may adjust the position from θ_n to θ_{n+1}. The corresponding trade, expressed in dollars, is

$$\Delta q_n = (\theta_{n+1} - \theta_n) S_n. \tag{3.5}$$

Trades incur proportional costs at rate λ:

$$\mathrm{TC}_n(\theta_n, \theta_{n+1}; S_n) = \lambda |\Delta q_n|. \tag{3.6}$$

The post-trade cash balance is

$$W_n^{\text{post}} = W_n - \Delta q_n - \mathrm{TC}_n(\theta_n, \theta_{n+1}; S_n), \tag{3.7}$$

which accrues interest at rate r until the next time step:

$$W_{n+1} = W_n^{\text{post}} e^{r\Delta t}. \tag{3.8}$$

The stock position θ_{n+1} carries over to t_{n+1} without further cost.

At maturity T, any remaining stock holdings are liquidated. Selling θ_N shares at price S_N yields proceeds $(1-\lambda)\theta_N S_N$, so that terminal cash wealth is

$$W_T = W_N + (1-\lambda)\theta_N S_N. \tag{3.9}$$

3.3 Tracking Error and Utility at Maturity

The European call payoff is

$$H(S_T) = \max(S_T - K, 0). \tag{3.10}$$

The hedger evaluates performance via the tracking error

$$\mathrm{TE} = W_T - H(S_T), \tag{3.11}$$

and seeks to maximize exponential utility

$$\mathbb{E}^Q\Big[-\exp\big(-\gamma\,\mathrm{TE}\big)\Big], \tag{3.12}$$

over all admissible trading strategies $\{\theta_n\}$.

3.4 Dynamic Programming Formulation

The state variables at time t_n are (S_n, W_n, θ_n). Let $V_n(S, W, \theta)$ denote the value function,

$$V_n(S, W, \theta) = \sup_{\{\theta_k\}_{k\geq n}} \mathbb{E}^Q_{n,S,W,\theta}\Big[-\exp\big(-\gamma\,\mathrm{TE}\big)\Big], \tag{3.13}$$

where the expectation is conditional on the state at t_n.

Terminal condition. At maturity $n = N$,

$$W_T = W_N + (1-\lambda)\theta_N S_N, \tag{3.14}$$

so

$$V_N(S_N, W_N, \theta_N) = -\exp\left(-\gamma\left(W_N + (1-\lambda)\theta_N S_N - H(S_N)\right)\right). \tag{3.15}$$

Bellman recursion. At time t_n with state (S_n, W_n, θ_n), choosing the next-period position θ_{n+1} implies:

$$W' = \big(W_n - (\theta_{n+1} - \theta_n)S_n - \lambda\big|(\theta_{n+1} - \theta_n)S_n\big|\big)\, e^{r\Delta t}. \tag{3.16}$$

Then the stock evolves to

$$S_{n+1} = \begin{cases} uS_n & \text{with probability } p, \\ dS_n & \text{with probability } 1-p. \end{cases} \tag{3.17}$$

The Bellman equation is

$$V_n(S_n, W_n, \theta_n) = \sup_{\theta_{n+1}} \left\{p\, V_{n+1}(uS_n, W', \theta_{n+1}) + (1-p)\, V_{n+1}(dS_n, W', \theta_{n+1})\right\}. \tag{3.18}$$

3.5 Numerical Implementation

State and Control Discretization

The CRR tree has $N = 220$ time steps and $n + 1$ nodes at level n. We index nodes by (n, j), where j is the number of up-moves. At node (n, j), the stock price is $S_{n,j} = S_0 u^j d^{n-j}$.

We discretize the number of shares θ on a grid

$$\theta \in \{\theta^{(1)}, \dots, \theta^{(N_\theta)}\}, \tag{3.19}$$

typically symmetric around zero (small short to small long). In practice, we choose a grid that covers the range of deltas encountered in the BSM model for the parameters considered. The wealth dimension is handled implicitly via the recursion; we restrict controls so that W_n remains within a range where the exponentials are numerically stable.

Backward Induction

The dynamic program is solved by backward induction:

1. Initialize $V_N(S_{N,j}, W, \theta)$ at maturity using the terminal condition over the discretized wealth and position grids.
2. For $n = N - 1, \dots, 0$:
   (a) For each stock node (n, j) and each admissible state (W, θ), enumerate candidate controls θ' on the grid.
   (b) For each θ', compute W' and the continuation value
   $$\mathrm{Cont}(\theta') = p\, V_{n+1}(S_{n+1,j+1}, W', \theta') + (1 - p)\, V_{n+1}(S_{n+1,j}, W', \theta'). \tag{3.20}$$
   (c) Select the maximizing control
   $$\theta^*_{n+1}(S_{n,j}, W, \theta) = \arg\max_{\theta'} \mathrm{Cont}(\theta'),$$
   and set
   $$V_n(S_{n,j}, W, \theta) = \max_{\theta'} \mathrm{Cont}(\theta').$$

The result is both a value function and an optimal feedback policy $\theta^*_{n+1} = \pi_n(S_n, W_n, \theta_n)$.

3.6 Indifference Pricing

Let $U_{\text{sell}}(W_0)$ denote the maximal expected utility when the market maker sells the call at premium W_0 at $t = 0$ and follows the optimal hedging strategy. Let $U_{\text{no trade}}(W_0)$ be the utility from not selling the option and simply investing W_0 in the money market until T.

Under exponential utility and risk-neutral dynamics,

$$U_{\text{no trade}}(W_0) = -\exp\left(-\gamma\, W_0 e^{rT}\right). \tag{3.21}$$

The indifference selling price $W_0^\star$ is defined implicitly by

$$U_{\text{sell}}(W_0^\star) = U_{\text{no trade}}(W_0^\star). \tag{3.22}$$

Numerically, $U_{\text{sell}}(W_0)$ is obtained from the DP value function at $n = 0$,

$$U_{\text{sell}}(W_0) = V_0(S_0, W_0, 0), \tag{3.23}$$

and $W_0^\star$ is found by root-finding on $U_{\text{sell}}(W_0) - U_{\text{no trade}}(W_0)$.

We then define the implied volatility σimpl as the volatility parameter that matches the indifference price in the frictionless BSM formula:

$$C_{\text{BS}}(S_0, K, T, r, \sigma\text{impl}) = W_0^\star. \tag{3.24}$$

Chapter 4

Optimal Hedging Strategy Results

4.1 Optimal Hedging Strategy and No-Trade Region

The dynamic program yields an optimal feedback policy

$$\theta_n^* = \pi_n(S_n, W_n, \theta_n), \tag{4.1}$$

which specifies, at each trading time t_n, how many shares to hold as a function of the current stock price, current position and cash wealth.

Because we use exponential utility with constant absolute risk aversion and proportional transaction costs, the optimal policy is *independent of the initial capital* W_0: changing W_0 only shifts the value function, but does not change the trading rule. Once the policy π_n has been computed by dynamic programming, it can therefore be applied to any starting wealth level.

No-Trade Region

A convenient way to visualize the policy is to look, at a fixed time t_n, at the decision

$$\theta_{n+1}^*(S_n, \theta_n)$$

for different combinations of current stock price S_n and current position θ_n. Figure 4.1 plots, for an intermediate time step, the region in the (S_n, θ_n) plane where the optimal decision is to *do nothing*, i.e. to keep $\theta_{n+1} = \theta_n$.

We observe a clear *no-trade band*: for each stock price S_n, there is an interval

$$[\theta_n^-(S_n), \theta_n^+(S_n)]$$

such that if the current position θ_n lies inside this band, the optimal decision is to avoid trading and save transaction costs. Only when θ_n moves outside this interval does the hedger adjust the position back toward the middle of the band. The existence of this region is the main qualitative difference with the frictionless, continuously rebalanced delta hedge.

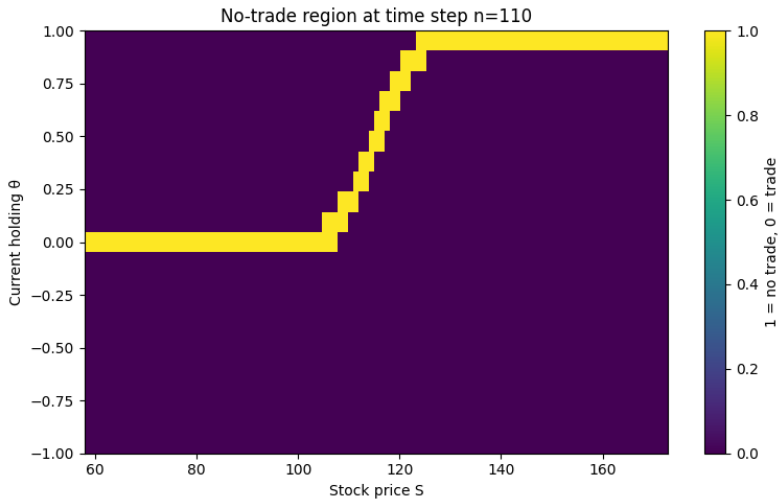

Figure 4.1: No-trade region for $\gamma = 1$ and $\lambda = 0.0005$ at an intermediate time ($n = 110$). The horizontal axis shows the stock price S_n along the tree, the vertical axis the current position θ_n. Yellow cells indicate states where no trade is optimal ($\theta_{n+1} = \theta_n$); dark cells correspond to states where a trade is executed.

Optimal Target Holdings vs. Black–Scholes Delta

To connect the optimal policy to familiar hedging intuition, we can look at the *target holding* as a function of the stock price when we start from a flat position ($\theta_n = 0$). Figure 4.2 shows $\theta_n^*(S_n, W_n, \theta_n = 0)$ for three different times to maturity and compares it to the standard Black–Scholes delta.

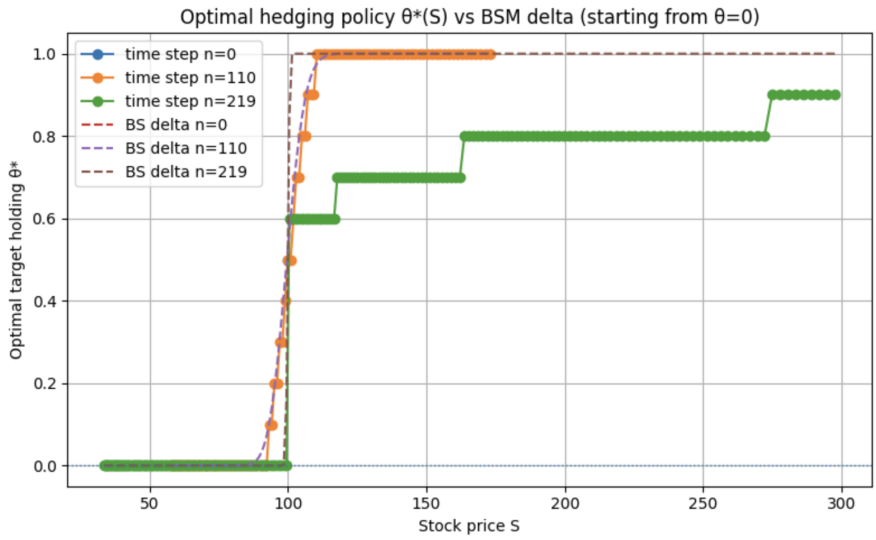

Figure 4.2: Optimal target holding $\theta_n^*(S_n, W_n, \theta_n = 0)$ as a function of the stock price for three time steps ($n = 0$, 110, and 219), compared with the corresponding Black–Scholes deltas.

Several patterns emerge:

- For most stock prices and away from maturity, the optimal target holding closely tracks the Black–Scholes delta, confirming that the frictionless delta is the natural reference point for hedging.

- Around the at-the-money region, the optimal holdings adjust in discrete steps rather than continuously. This reflects both the discretization of the θ grid and the fact that small changes in delta are not worth trading for when transaction costs are present.
- As maturity approaches and the option becomes deep in or out of the money, the optimal position saturates near 0 or 1, again broadly in line with the delta.

Together, Figures 4.1 and 4.2 highlight the main qualitative features of the optimal hedging strategy: it behaves like a discretized version of the Black–Scholes delta, but with an endogenous no-trade region that prevents excessive rebalancing when trading is costly.

4.2 Indifference Price and Implied Volatility

For the baseline at-the-money case ($K = 100$, $\gamma = 1$, $\lambda = 0.0005$), the frictionless BSM price with $\sigma = 0.25$ is approximately

$$C_{\mathrm{BS}} \approx 3.16.$$

Solving the DP and applying the indifference condition yields an indifference price around

$$W_0^\star \approx 3.34.$$

Inverting the BSM formula, this corresponds to an implied volatility σ_{impl} of about 26.5%, so that the presence of transaction costs and risk aversion effectively shifts the volatility parameter by roughly 1.5 percentage points.

Table 4.1: Indifference price and implied volatility for the at-the-money call.

Model	Price	Implied Volatility	Difference vs. BSM
BSM (no transaction costs)	$C_{\mathrm{BS}} \approx 3.16$	$\sigma = 0.25$	–
Indifference ($\lambda = 0.0005$)	$W_0^\star \approx 3.34$	$\sigma_{\mathrm{impl}} \approx 0.265$	≈ 0.18

This gap between $W_0^\star$ and C_{BS} reflects both the expected transaction costs and the residual hedging risk borne by the risk-averse market maker.

4.3 Properties of the Optimal Hedging Strategy

4.3.1 Distribution of Tracking Error under Optimal Hedging

We simulate M paths of the CRR tree under Q and apply the optimal policy along each path. Starting from an initial premium equal to the frictionless BSM price, we compute the realized tracking error

$$\mathrm{TE}^{(m)} = W_T^{(m)} - H(S_T^{(m)}),$$

for each path m.

In our simulations, the tracking-error distribution is centered close to zero but not perfectly symmetric. A representative run yields a sample mean and standard deviation approximately

$$\bar{\mathrm{TE}} \approx 0.059, \qquad \widehat{\sigma}_{\mathrm{TE}} \approx 0.347.$$

The slight positive mean reflects the fact that, starting from the frictionless price, the optimal hedger tends to over-hedge a bit in order to reduce downside risk, while transaction costs are partly offset by favorable hedging outcomes on some paths. The distribution exhibits somewhat heavier tails than a Gaussian with the same mean and variance, due to rare large stock movements combined with finite trading frequency.

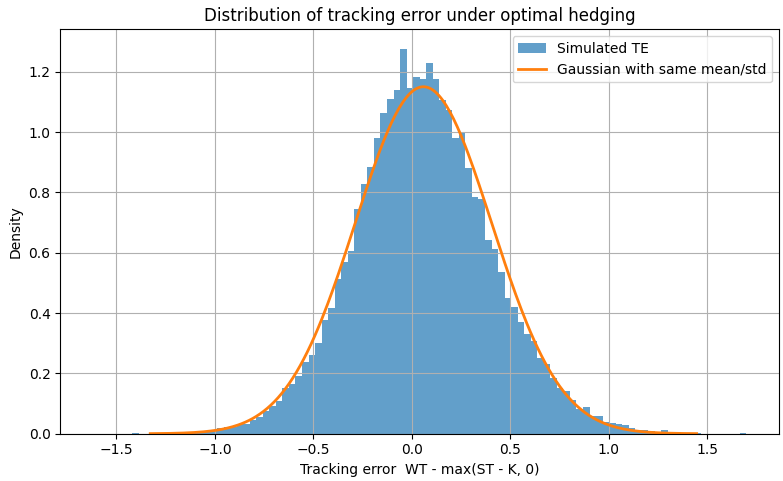

Figure 4.3: Distribution of tracking error under the optimal strategy for $\gamma = 1$ and $\lambda = 0.0005$.

4.3.2 Risk Aversion and the Mean–Variance Trade-off

To explore the effect of risk aversion, we repeat the DP and simulation for several values of γ (for example, $\gamma \in \{0.5, 1, 2\}$). For each γ, we keep the initial premium at the frictionless BSM price and compute the mean and standard deviation of TE under the optimal policy.

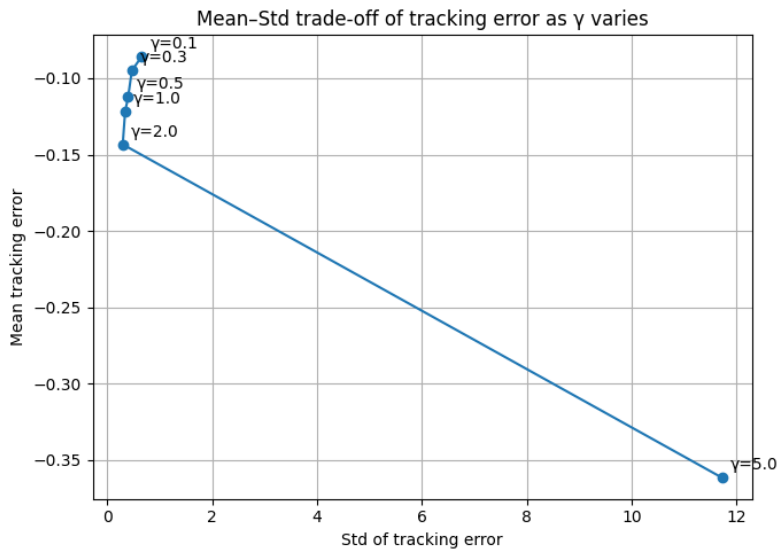

Figure 4.4: Mean and standard deviation of tracking error under the optimal strategy as a function of risk aversion γ. Each point corresponds to a different value of γ.

The pattern is the following:

- For small γ, the hedger is nearly risk-neutral with respect to tracking error and is willing to accept large dispersion in TE to save on transaction costs. This results in a relatively wide tracking-error distribution.
- As γ increases, the hedger trades more frequently and keeps θ_n closer to the frictionless

delta, thereby reducing the magnitude of extreme tracking errors. The mean TE tends to move closer to zero, but total transaction costs increase.

- For very high γ, the strategy becomes very conservative: the hedger attempts to closely match the payoff even at the cost of frequent trading, which would eventually push the indifference price further above the frictionless value.

These trade-offs are clearly visible in Figure 4.4.

4.3.3 Trading Frequency and No-Trade Durations

The optimal policy does not rebalance at every step. To quantify this, we record the number of time steps between successive trades along each simulated path and plot the empirical distribution of these *no-trade durations*.

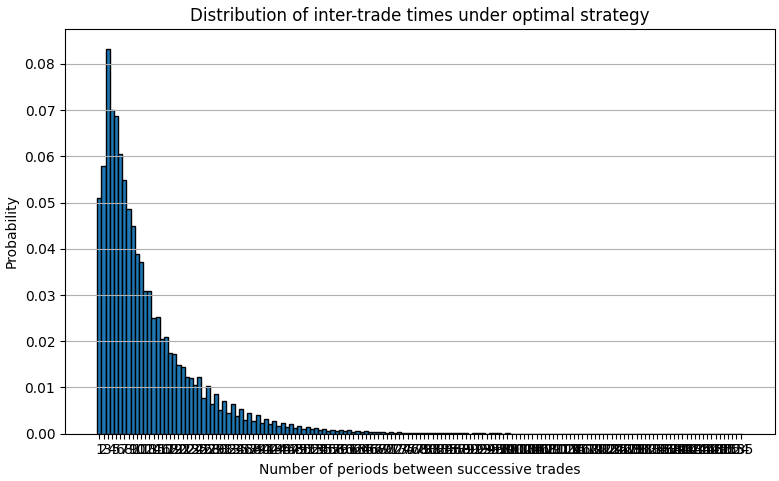

Figure 4.5: Distribution of the number of time steps between successive trades under the optimal strategy.

Most no-trade periods are short (one to a few time steps), reflecting the relatively high trading frequency in our discrete-time approximation. However, a non-negligible fraction of paths exhibit longer periods with no trades, particularly when S_n evolves in regions where the current hedge lies inside the no-trade band. This confirms that the optimal policy endogenously generates state-dependent trading schedules, rather than a fixed calendar.

Chapter 5

Heuristic Black–Scholes Delta Hedging

5.1 Heuristic Delta-Hedging Strategy

As a benchmark, we consider the following Black–Scholes delta-hedging :

- At fixed rebalancing times $t_0, t_k, t_{2k}, \ldots$, compute the Black–Scholes delta of the call,

$$\Delta_{\mathrm{BS}}(S_t, t; K, T, \sigma, r),$$

  under the frictionless BSM model with volatility σ.

- Adjust the stock position to $\theta_t = \Delta_{\mathrm{BS}}$ at those times only.
- Between rebalancings, hold θ_t constant; trades incur proportional costs at rate λ.

We simulate this strategy for several rebalancing intervals (binomial time steps) :

$$k \in \{1, 2, 5, 10, 20, 50\}$$

and, for each k, compute the distribution of tracking error TE as well as the corresponding expected utility $\mathbb{E}[-\exp(-\gamma \mathrm{TE})]$.

5.2 Comparison with the Daily Rebalancing

A natural practical benchmark is "once per day" rebalancing. Since our model has 10 trading times per day, this corresponds to $k = 10$. Figure 5.1 compares the distribution of tracking error under the optimal dynamic-programming policy and under the heuristic strategy with daily rebalancing.

Both distributions are roughly centered near zero, but the heuristic hedge has visibly fatter tails: extreme tracking errors are more frequent. This is consistent with the fact that the heuristic trades on a fixed calendar schedule, regardless of whether the current hedge is far from or close to the desired exposure, whereas the optimal policy trades only when the state leaves the no-trade region. In expected-utility terms, the daily heuristic delivers a lower value than the optimal strategy, reflecting this higher tail risk.

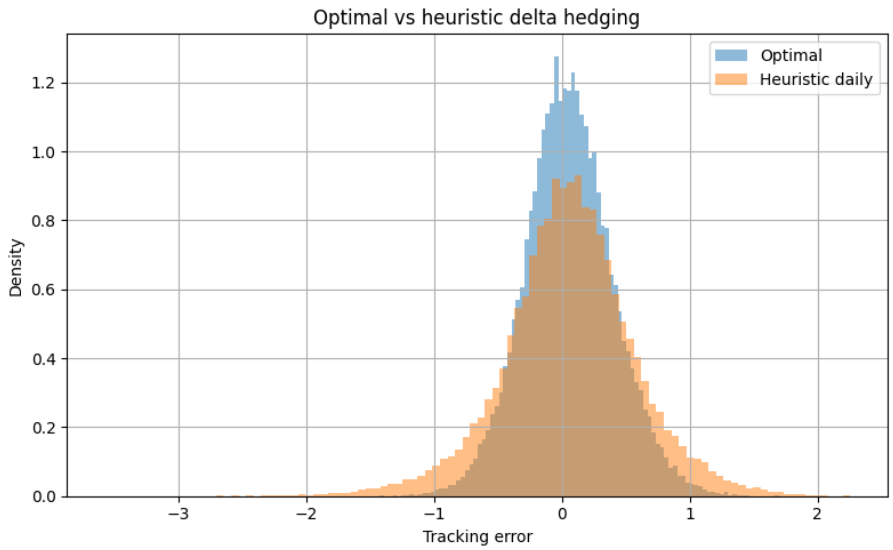

Figure 5.1: Distribution of tracking error under the optimal strategy and under the heuristic strategy with daily rebalancing ($k = 10$).

5.3 Choosing the Rebalancing Frequency

We now treat the rebalancing interval k as a design parameter and search for the value that makes the heuristic strategy as attractive as possible. Table 5.1 reports the mean tracking error, its standard deviation, and the expected utility for each k.

Table 5.1: Heuristic delta-hedging performance for different rebalancing frequencies k.

Rebalancing every k steps	Mean TE	Std. TE	Expected Utility
$k = 1$	-0.1645	0.1078	-1.1857
$k = 2$	-0.0223	0.1996	-1.0434
$k = 5$	0.0224	0.3633	-1.0479
$k = 10$	0.0652	0.5195	-1.0827
$k = 20$	0.0829	0.7482	-1.2762
$k = 50$	0.1070	1.0866	-2.0410

To visualize the trade-off, Figure 5.2 plots the mean tracking error against its standard deviation for each k.

The figure and the table together show a clear pattern:

- For very frequent rebalancing ($k = 1$), the standard deviation of TE is small, but the mean TE is strongly negative and expected utility is poor. The hedge is accurate, but transaction costs are extremely high.
- For very infrequent rebalancing ($k = 50$), transaction costs are low, but hedging errors are very volatile and expected utility again deteriorates.
- Intermediate values of k strike a better balance. Among the values tested, $k = 2$ **maximizes expected utility**: it keeps the mean tracking error close to zero while keeping dispersion at a moderate level.

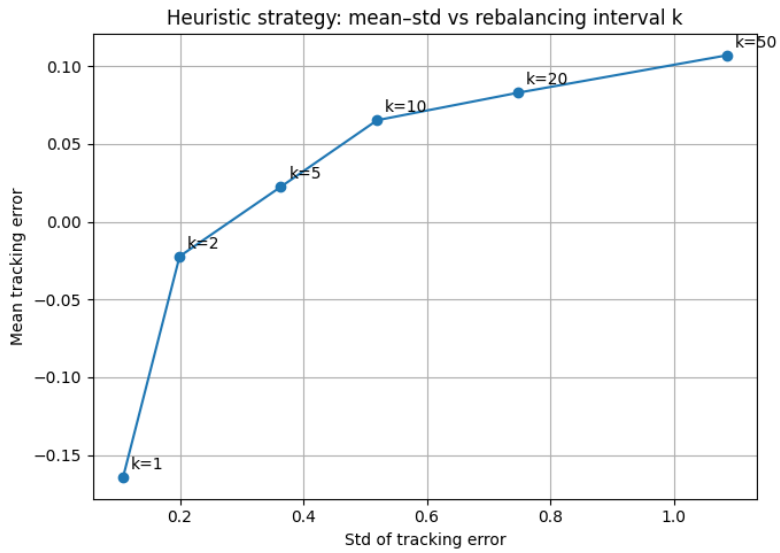

Figure 5.2: Heuristic strategy: mean–standard deviation trade-off as a function of the rebalancing interval k. Each point is labelled by its k value.

The performance of the heuristic strategy exhibits a clear trade-off between transaction costs and hedging accuracy, which is illustrated in Figure **??**. Very frequent rebalancing ($k = 1$) produces a low dispersion of tracking error but a strongly negative mean, indicating that excessive trading costs dominate the hedging benefits. In contrast, very infrequent rebalancing ($k = 20$ or $k = 50$) drastically reduces transaction costs but results in large unhedged exposure and high volatility of terminal error.

An intermediate rebalancing frequency, around $k = 2$ to $k = 5$, achieves the best compromise. In this region, the mean tracking error is close to zero while the standard deviation remains moderate, leading to the highest expected utility among the heuristic strategies. This optimal k balances two opposing effects:

- reducing transaction costs by avoiding over-trading,
- while still controlling exposure to large price movements.

We therefore take $k^* = 2$ as the *optimal rebalancing frequency* within this class of heuristic strategies.

Summary of Heuristic vs. Optimal Hedging :

Relative to the dynamic-programming policy, the Black–Scholes delta heuristic:

- trades on a fixed time grid rather than when the state leaves the no-trade region,
- is highly sensitive to the choice of rebalancing interval k, and
- delivers strictly lower expected utility for all tested k, even at the best frequency $k^* = 2$.

Nevertheless, with a well-chosen rebalancing frequency, the heuristic can get reasonably close to the optimal strategy, especially for moderate transaction costs and short maturities. This helps to rationalize the widespread use of simple delta-hedging rules in practice, while also quantifying the performance gap relative to a fully optimized, state-dependent hedging policy.

Chapter 6

Parameter Sensitivity

6.1 Out-of-the-Money Strikes

We extend the analysis to out-of-the-money options with strikes $K > 100$, specifically $K = 110$ and $K = 120$. For each strike and each transaction cost level λ, we solve the dynamic programming problem, compute the indifference price $W_0^\star(K, \lambda)$, and analyze the resulting hedging errors.

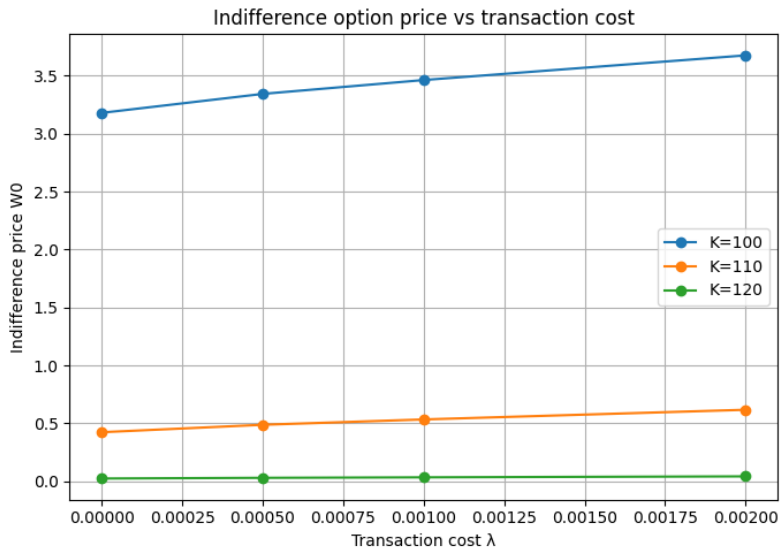

Figure 6.1: Indifference option price $W_0^\star$ as a function of transaction cost λ for different strikes K.

For any fixed transaction cost, the indifference price decreases as the strike increases, reflecting the lower intrinsic value and reduced probability of exercise for more OTM options.

However, the impact of transaction costs differs significantly across strikes. While absolute price levels are small for deep out-of-the-money options, the proportional impact of transaction costs remains meaningful. This is due to the fact that even infrequent hedging trades remain costly relative to the small option premium.

Moreover, at-the-money options remain the most sensitive to transaction costs because of their high gamma, which generates frequent rebalancing needs. In contrast, deep out-of-the-money options require very few trades, making both their transaction costs and their hedging risks significantly smaller.

6.2 Impact of the Transaction Cost Parameter λ

We now fix the strike and vary the transaction cost parameter over

$$\lambda \in \{0,\ 0.0005,\ 0.001,\ 0.002\}.$$

For each value of λ, we solve the dynamic program and compute the indifference price, simulate the optimal hedging strategy, compute the mean and standard deviation of the tracking error, and measure the average number of trades per path.

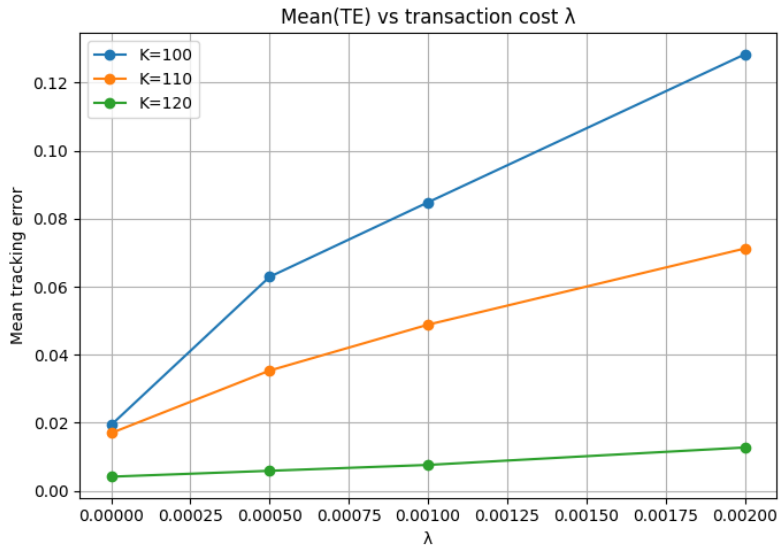

Figure 6.2: Mean tracking error as a function of transaction cost λ.

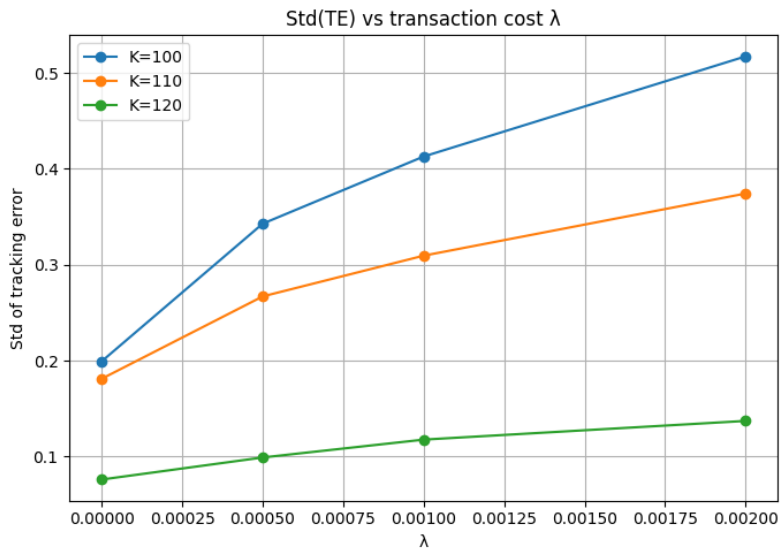

Figure 6.3: Standard deviation of tracking error as a function of transaction cost λ.

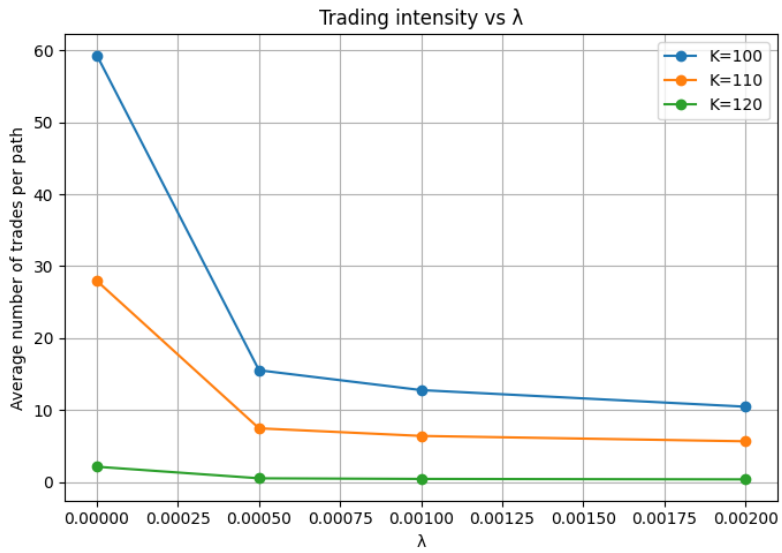

Figure 6.4: Average number of trades per path as a function of transaction cost λ.

Several robust effects emerge from these results:

- **Increase in tracking error:** Both the mean and the standard deviation of the tracking error rise monotonically with λ for all strikes. Higher transaction costs discourage frequent rebalancing, leading to larger residual hedging mismatches at maturity.
- **Endogenous reduction in trading activity:** Figure 6.4 shows that the average number of trades per path collapses rapidly as λ increases. This reflects the widening of the optimal no-trade region: the hedger tolerates increasingly large deviations from the frictionless hedge before intervening.
- **Strike dependence:** At-the-money options exhibit the strongest sensitivity to transaction costs. They require frequent rebalancing because of high gamma, making both their trading intensity and tracking error volatility much larger than for out-of-the-money options. Deep out-of-the-money options trade very rarely and exhibit both low transaction costs and low hedging risk.
- **Increase in indifference prices:** As transaction costs increase, the indifference option price rises for all strikes. The market maker requires a higher premium to compensate for both the higher expected trading costs and the greater unhedgeable risk caused by reduced rebalancing.

Overall, transaction costs fundamentally reshape both the structure of the optimal hedging strategy and the distribution of terminal hedging errors. Higher costs induce wider no-trade regions, sharply lower trading frequency, larger tracking error dispersion, and higher equilibrium option prices. These effects are strongest for options near the money and become progressively weaker for far out-of-the-money contracts.

Chapter 7

Discussion and Limitations

7.1 Effects of Transaction Costs

Our analysis highlights how even small proportional costs can fundamentally alter hedging behavior:

- The classical continuous-time replication strategy is no longer optimal; instead, the hedger optimally tolerates deviations from the frictionless delta within a no-trade region.
- The effective hedging policy becomes a state-dependent free-boundary problem in discrete time, where boundaries depend on γ, λ, T, and K.
- The resulting option price is no longer pinned down by a single replication argument; instead, the indifference price depends on both market frictions and investor preferences.

7.2 Optimal Strategy vs. Heuristic Delta Hedging

The contrast between the dynamic-programming policy and the heuristic delta strategy can be summarized as follows:

- The heuristic delta hedge with fixed rebalancing interval k is *time-based*: it trades on a calendar schedule regardless of the state. The DP policy is *state-based*: it trades only when the current position leaves the no-trade region.
- For a given (γ, λ), the DP policy systematically dominates the heuristic in expected utility, and typically yields a more favorable mean–variance profile for TE.
- However, for moderate transaction costs, a well-chosen heuristic (e.g. $k = 2$) can deliver reasonably good performance, which helps to explain why such rules are often used in practice.

7.3 Limitations and Possible Extensions

Key limitations of our implementation include:

- **State-space discretization:** The DP relies on finitely many θ grid points and an implicit wealth grid, which introduces discretization error and limits resolution.

- **Single-asset setting:** Our analysis focuses on a single underlying stock and a single option. In practice, hedgers manage multi-asset portfolios and may use multiple hedging instruments.
- **Simplified cost structure:** We assume purely proportional costs with a constant rate λ. Realistic execution costs may have fixed components, nonlinear impact, and dependence on trading volume and market conditions.

Potential extensions include incorporating stochastic volatility or jumps in S_t, modeling more realistic cost functions, and using regression-based or machine-learning approximations to scale the DP method to higher dimensions.

Chapter 8

Conclusion

We developed and implemented a dynamic programming framework for optimal hedging of a short European call under proportional transaction costs and exponential utility. Using a high-frequency CRR approximation to the Black–Scholes–Merton model, we characterized the optimal hedging policy, documented the structure of the resulting no-trade region, and computed utility-based indifference prices and implied volatilities.

Simulations reveal that the optimal policy achieves a favorable compromise between hedging accuracy and cost minimization and clearly outperforms simple Black–Scholes delta-hedging rules in terms of expected utility and tracking-error performance. Parameter sensitivities with respect to risk aversion γ, transaction costs λ, and strike K were found to be economically intuitive and informative for understanding how micro-level trading frictions map into macro-level option prices.

Overall, the project shows how tools from stochastic control and numerical dynamic programming can be used in a realistic setting to analyze hedging under transaction costs and to derive consistent, utility-based option prices.

Chapter 9

Resources and Use of AI-Assisted Tools

We consulted course lecture notes and official documentation for our software stack (NumPy, SciPy, pandas, matplotlib, and statsmodels) for implementation details, numerical routines, and plotting conventions.

We used *AI-assisted tools* only to streamline programming tasks (e.g., refactoring, debugging, and syntax suggestions). All modeling ideas, derivations, calibrations, empirical analyses, and written exposition are our own.